MIL-P-7206B(USAF)
9 Nov 1970
SUPERSEDING
MIL-R-7206A(USAF)
18 August 1964

MILITARY SPECIFICATION

POWER SUPPLY, SILICON RECTIFIER, TYPE B-8

1. SCOPE

* 1.1 This specification covers one type of portable, 200-ampere, 28-volt direct current (DC) power supply designated Type B-8.

2. APPLICABLE DOCUMENTS

* 2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

PPP-B-601	Boxes, Wood, Cleated-Plywood

Military

MIL-P-116	Preservation, Methods Of
MIL-P-6457	Power Supply, Metallic Rectifier Type, General Specification For

STANDARDS

Military

MIL-STD-129	Marking For Shipment And Storage

Air Force-Navy Aeronautical Standards

AN2551	Cable Assembly, Power, Electrical

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

* 3.1 First article. This specification makes provisions for first article inspection.

3.2 General specification. The requirements of MIL-P-6457 apply as requirements of this specification with the exceptions and additions specified herein. When the two specifications conflict, this specification shall govern.

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3.3 Design and construction.

* 3.3.1 Configuration. The power supply shall be designed and constructed as a portable unit with a configuration as shown on Figure 1. Two rubber-tired wheels with a minimum diameter of 8 inches shall be provided. The load on each wheel shall not exceed its rated load limit.

* 3.3.1.1 Handle. A suitable handle shall be provided to enable personnel to maneuver the power supply. The handle shall not interfere with the power cables or instruments.

* 3.3.1.2 Cable racks. Cable racks shall be provided for mounting the alternating- and direct-current power cables and shall be designed to permit complete portability of the power supply.

* 3.3.1.3 Standard parts. AN or MIL standard parts shall be used where they suit the purpose and shall be identified by their part numbers. Commercial utility parts such as screws, bolts, nuts, et cetera, may be used provided they have suitable properties and are replaceable by the AN or MIL standard parts without alteration and provided the corresponding AN or MIL part numbers are referenced on the drawings and in the parts list. In applications for which no suitable corresponding AN or MIL part is in effect on the date of invitation for bids, commercial parts may be used provided written approval is obtained from the cognizant engineering activity. Use of nonapproved, nonstandard components will cause rejection of first article.

3.3.2 Electrical characteristics.

* 3.3.2.1 Input voltage. The power supply shall have an input voltage of 220/440 volts, 3 phase, 60 cycles. All controlled circuits, such as operating coils of starters, pushbuttons, and relays, shall operate on 110 volts or less. This may be accomplished by means of a small external control transformer that lowers the voltage for the control devices or by a separate winding on the power transformer.

* 3.3.2.2 Output rating. The continuous output rating of the power supply shall be 200 amperes at 28 volts.

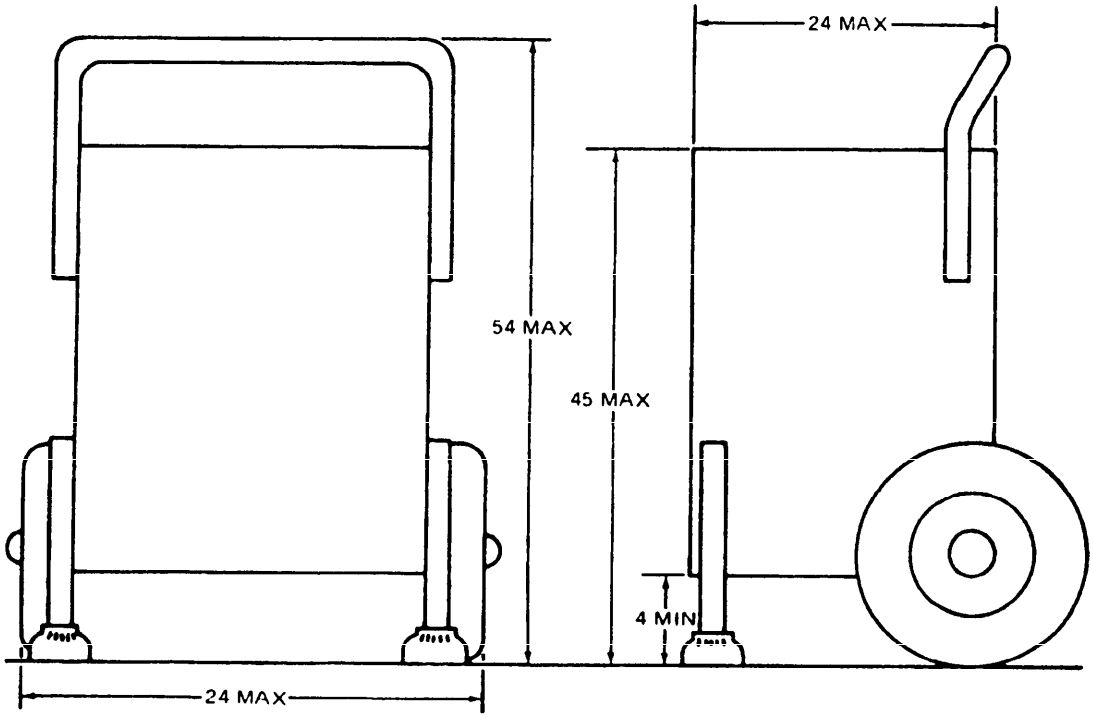
* 3.3.2.3 Efficiency. The efficiency of the power supply shall be not less than 75 percent.

3.3.2.4 Regulation. With the input voltage remaining constant at the nominal rated value, the no-load voltage at the output terminals shall remain within the range of 28.0 to 30.5 volts. The output voltage shall vary not more than 2.5 volts as the load is changed from no load to full load.

* 3.3.2.5 Ripple voltage. The peak alternating-current ripple voltage superimposed on the direct-current voltage output shall not exceed 1.2 volt peak when determined for both polarities.

* 3.3.2.6 Bleeder resistor. If the power supply design requires a bleeder resistor, the resistor shall be installed in a manner to dissipate the generated heat by means of the airstream flowing through the unit.

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DIMENSIONS IN INCHES

FIGURE 1. POWER SUPPLY CONFIGURATION

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* 3.3.2.7 Rectifiers. The rectifiers shall be silicon diodes and shall be connected for full-wave rectification. When the power supply is operated continuously in an ambient temperature of 55° Centigrade (C) at full rated load, the temperature of the diodes shall not exceed 120°C.

* 3.3.2.7.1 Rectifier elements. Rectifiers elements shall be of the silicon type.

* 3.3.2.8 Input circut breaker. A 220/440 volt alternating current, time delay, magnetic-hydraulic circuit breaker shall be provided in the input circuit for overload and starting purposes. Without tripping, the circuit breaker shall meet the surge rating tests and the intermittent duty tests as specified in Section 4 of MIL-P-6457. In addition, when an overload of 400 percent is applied, the circuit breaker shall trip before any damage is incurred by the unit.

3.4 Power cables.

* 3.4.1 Input cable. The power supply shall be provided with a 4-wire permanently attached input cable. The cable shall be type I, 100 feet long, and shall have a maximum line voltage drop of 2 percent rated voltage at continuous rated load.

* 3.4.2 Output cables. The power supply shall be provided with two cable assemblies conforming to AN2551D20.

3.5 Meters. Commercial-type meters with a 2-percent accuracy shall be provided.

3.5.1 Ammeter. The scale range of the direct-current ammeter shall be 0 to 300 amperes.

3.5.2 Voltmeter. The scale range of the direct-current output voltmeter shall be 0 to 40 volts direct-current.

* 3.6 Dimensions. The dimensions of the power supply shall be in accordance with Figure 1.

* 3.7 Weight. The total net weight of the power supply, exclusive of cables, shall not exceed 325 pounds.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.2 Inspection and testing. The power supply shall be inspected and tested in accordance with MIL-P-6457 and the additional first article inspections specified in 4.3.

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* 4.3 Test methods.

* 4.3.1 Silicon rectifier temperature rise. The power supply shall be operated at rated load in ambient temperature of 55°C until thermal equilibrium is reached. Thermocouples shall be placed in at least three places in the silicon diodes to determine that the temperature of the diodes does not exceed 120°C. If desired, this test may be combined with the high temperature test specified in 4.6.15a of MIL-P-6457.

* 4.3.2 Input circuit breaker test. The power supply shall be operated in an ambient temperature of 25°C with an input of 220 volts alternating current and then reconnected for 440 volt alternating current operation. It shall be shown that with each input voltage, when a 400 percent load is applied, that the input circuit breaker will trip before any damage is incurred by the unit. Immediately at the conlcusion of this test, the power supply shall meet the regulation and efficiency tests as specified in MIL-P-6457. If desired, this test may be combined with the operational test specified in MIL-P-6457.

5. PREPARATION FOR DELIVERY

* 5.1 Preservation and packaging. Preservation and packaging shall be Level A or C as specified (see 6.2).

* 5.1.1 Level A. Each power supply shall be preserved and packaged Method IIa of MIL-P-116. Each power supply shall be placed in a wood cleated-plywood unit container conforming to PPP-B-601, type shall be as specified in 5.2.

* 5.1.2 Level C. Each power supply shall be preserved and packaged in a manner which will afford protection against corrosion, deterioration, and physical damage during shipment from the supply source to the first receiving activity for immediate use. This level may conform to the supplier's commercial practice when such meets the requirements of this level.

* 5.2 Packing. Packing shall be Level A, B or C as specified (see 6.2).

* 5.2.1 Level A. The unit container specified in 5.1.1 shall be the shipping container and shall conform to PPP-B-601, overseas type. Containers shall be of minimum cube and tare consistent with the protection required. Container closure and strapping shall be in accordance with the appendix to PPP-B-601.

* 5.2.2 Level B. Level B shall be the same as Level A except containers shall conform to domestic type.

* 5.2.3 Level C. Each power supply shall be packed in a manner which affords adequate protection against damage during direct shipment from the supply source to the first receiving activity for immediate use. This level shall conform to applicable carrier rules and regulations and may be the supplier's commercial practice when such meets the requirements of this level.

* 5.3 Marking. In addition to any special marking required by the contract or order, containers shall be marked in accordance with the requirements of MIL-STD-129.

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6. NOTES

* 6.1 Intended use. The type B-8 power supply is intended for use on the ground to supply electrical energy to aircraft with 28 volts direct current electrical systems.

* 6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.

b. Selection of applicable level of preservation, packaging and packing (see 5.1 and 5.2).

* 6.3 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Air Force - 82

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Project No. 6130-F102

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

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REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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